THE CLOCKWISE PRIOR: SILENT, DIRECTED CONVENTION REVERSION IN LANGUAGE MODELS

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ABSTRACT

When a language model is instructed to use a sign convention that conflicts with the convention dominant in its training text, what does it do? Across three exactly-verified physics and mathematics domains (mesh-current circuit analysis, oriented line integrals, and three-dimensional vector mechanics), eight instructable conventions, and fifteen model configurations spanning a $\sim$790$\times$ competence range, we find that sign-convention failures are *silent and directed*: among 49,014 magnitude-correct diagnostic variables, 99.85% of all sign errors reproduce the trained convention's value exactly, and on the seven pure sign-flip conventions the rate is 100%. Whether reversion occurs is governed by three measurable factors. (1) *Reversal pedagogy*: conventions whose reversal is absent from textbooks (clockwise mesh currents) defeat every frontier model (76.0% reversion), while conventions whose reversal is itself taught (clockwise contour orientation, left-hand cross products) are obeyed ($\leq$2.3–0.2%); a weighted-logistic trend across all eight conventions gives $z = 45.4$. (2) *Cognitive load*: within a taught-reversal domain, raising task difficulty reactivates the prior dose-dependently (1.0%$\rightarrow$35.8% as strain rises), while a transform-matched control requiring the *identical* output edit stays $\leq$0.4–0.3%, isolating convention entrenchment from generic instruction loss. (3) *Capability*: mid-scale models revert on even taught-reversal conventions (91–99%) that frontier models obey. Two mitigations fail: enabling reasoning raises competence by up to 76 points while moving compliance by 2–4 points (five model scales, two domains), and in-context worked demonstrations of the required convention leave reversion essentially unchanged in two of three frontier models (GPT-5: 70.6%$\rightarrow$73.2% after three examples). Two pre-registered predictions were refuted and are reported as such. All answers are graded sign-exactly against four-way independent oracles; the benchmark regenerates deterministically by seed.

1 INTRODUCTION

A language model asked to analyze a circuit with mesh currents defined counter-clockwise will often do everything right—set up the equations, solve the system, recover every magnitude—and then report the signs a clockwise analysis would have produced. The answer looks correct: units, structure, and magnitudes are all right. Only a sign-exact check against the instructed convention reveals that the model has silently substituted the convention it was trained on for the one it was told to use. We call this failure mode *convention reversion*, and this paper measures it, maps what governs it, and tests what fails to fix it.

Sign conventions are an unusually clean probe of instruction following. They are *arbitrary* (both choices are physically valid, so the instruction is legitimate), *binary* (compliance is a sign), and *exactly gradable* (a solver oracle can compute the correct answer under either convention). Because magnitude and sign are separately gradable, competence and compliance can be measured independently within a single response—avoiding the confound in which instruction-following failures are mistaken for capability failures, and vice versa.

Contributions. (1) *An instrument*: three procedurally generated domains (Section 2) in which every answer is verified by four independent solvers (three exact, zero tolerance), every convention transform is independently re-realized (e.g., a left-handed convention is validated by re-solving with the Levi-Civita symbol negated), and grading is fully deterministic—no LLM judges anywhere. (2) *A phenomenon*: reversion is silent and directed. 99.85% of 49,014 sign errors match the trained frame exactly; the residual 0.15% occurs only under the study's single affine (non-sign-flip) transform (Section 3). (3) *An anatomy*: reversion is governed by reversal pedagogy, load, and capability (Section 4), including a dose-response with a transform-matched control that isolates entrenchment causally. (4) *Two negative results on mitigation*: reasoning modes and in-context worked demonstrations both fail to restore compliance (Section 5). (5) *A falsification record*: two predictions pre-registered in the project history were refuted by the data and are reported as refuted (Section 6).

2 THE INSTRUMENT

Domains. *Circuits*: linear DC networks across seven topology families (supermesh, dependent sources, bridges, ladders); requested variables are mesh currents, node voltages, and branch quantities. *Contour*: oriented line integrals of polynomial vector fields around lattice polygons (including non-convex); requested variables are per-edge work integrals, total circulation, and flux. *Vectors*: multi-body torque, angular-momentum, and magnetic-force systems in $\mathbb{R}^3$; requested variables are vector components and invariant scalars. Difficulty is a config knob (element count, polynomial degree, coordinate magnitude); each domain ships an easy and a hardened tier.

Conventions as contracts. Each instance is posed under a *convention contract*: the default cell states the textbook convention explicitly; flip cells change exactly one convention (e.g., "mesh currents are defined COUNTER-CLOCKWISE"). Eight conventions are instructed across the domains (Table 1). Every variable carries a role determining which contracts flip it, giving per-cell diagnostic masks; a solution-method contract (e.g., mesh vs. nodal analysis; direct vs. transfer theorem) separates reporting-frame effects from procedure effects. Critically, two contour cells—clockwise orientation (`cw`) and work-done-against-the-field (`wrk`)—demand the *byte-identical* output transform (negate all work values) while naming different conventions: any behavioral difference between them isolates convention identity with output demands perfectly matched.

Verification. Every accepted instance passes a four-way oracle: in circuits, two independent hand-derived formulations, an exact symbolic MNA solver, and NGSPICE; in contour, symbolic parametrized integration, Green's-theorem double integration over an exact triangulation, an independent rational-arithmetic engine, and numerical quadrature; in vectors, component formulas, Levi-Civita summation, symbolic cross products, and floating-point evaluation, plus exact transfer-theorem identities. Three of four routes are exact rational arithmetic compared at zero tolerance; datasets report `inconsistent` = 0 by construction. Expected answers under every contract are additionally validated by *independent physical realization*: the clockwise cell by re-grounded re-simulation, the left-handed cell by re-solving with $\epsilon_{ijk} \to -\epsilon_{ijk}$, the reaction cell by negating force sources (125/125 physics PASS in each domain). Grading is deterministic: a mandated answer line is parsed by rules; a prediction is *reverted* only if it matches the trained-frame value within tolerance where the instructed frame demanded the opposite sign.

Panel. Fifteen configurations: GPT-5 (low reasoning effort), gpt-oss-120b, Gemma-4-31B, DeepSeek-V3, Llama-3.3-70B, Qwen2.5-72B, and the Qwen3 dense ladder (1.7B–32B) with its native reasoning toggle, spanning $\sim$790$\times$ in whole-problem competence. Decoding policy is uniform; provenance (serving stack, quantization) is pinned and recorded per row.

3 REVERSION IS SILENT AND DIRECTED

Across all domains, tiers, cells, and models we grade 49,014 magnitude-correct diagnostic variables. 99.85% of all sign errors equal the trained convention's value exactly; 74 incoherent cases (0.15%) occur, and every one falls in the circuits reference-node cell—the study's only *affine* transform, where partially applying the potential shift produces a value matching neither frame. On all seven pure sign-flip conventions, the incoherence count is zero. Sign errors in these models are not noise: they are the training distribution showing through, variable by variable.

The paired within-physics design makes the effect causal. In circuits, the same physics posed under the default vs. counter-clockwise contract yields aggregate discordant counts b=401 (correct at default, reverted at flip) vs. c=6 across the panel, with exact McNemar $q_{\mathrm{BH}} \leq 0.006$ in every

Table 1: The entrenchment gradient: frontier low-strain reversion by reversal-pedagogy tier (pooled over GPT-5, gpt-oss-120b, Gemma-4-31B; Wilson 95% CIs).

Convention (domain)	Tier	Reversion %	95% CI	n vars
mesh direction (circuits)	2	76.0	[73.3, 78.5]	1,045
passive sign (circuits)	1	6.1	[4.6, 8.0]	755
reference node (circuits)	1	0.5	[0.1, 1.7]	437
handedness (vectors)	1	0.2	[0.1, 0.4]	3,610
orientation (contour)	0	2.3	[1.9, 2.8]	3,902
reaction pair (vectors)	0	1.5	[1.1, 1.9]	3,570
flux normal (contour)	0	0.1	[0.0, 0.8]	736
work sign (contour)	0	0.0	[0.0, 0.1]	4,023

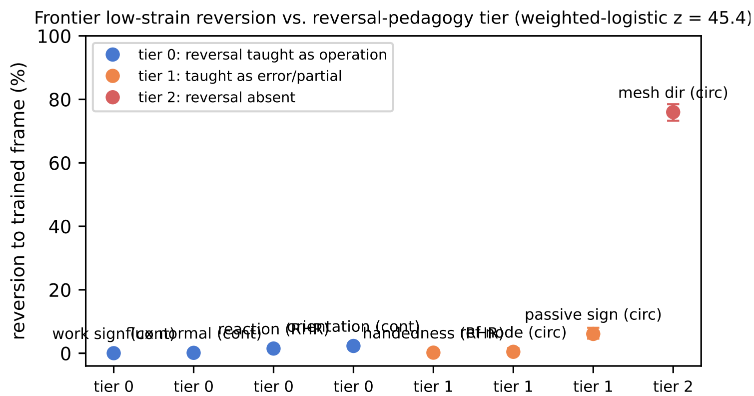

Figure 1: Reversion vs. reversal-pedagogy tier across all eight conventions.

competent family. Competence does not immunize: Gemma-4-31B solves 85% of circuit magnitudes yet reverts 77.9% of diagnostic signs. The same models comply at 85–97% with matched *non-convention* unusual instructions, and test–retest resampling (k=5) bounds run-to-run variation at a few points—the rate is a stable property, not sampling luck.

4 What governs reversion

4.1 Reversal pedagogy: the entrenchment gradient

The eight conventions differ sharply in whether their *reversal* appears in training-adjacent text. We code each on an ordinal pedagogy tier from a documented audit of standard textbooks: tier 0, the reversal is taught as a routine operation (reversing a contour's orientation negates the integral—a theorem in every calculus text; reaction forces; inward normals); tier 1, the reversal appears only as an error-contrast or partial treatment (the left hand gives the opposite direction; active sign convention; non-ground references); tier 2, the reversal is essentially absent (no circuits textbook we audited works a counter-clockwise mesh example). Table 1 and Figure 1 give the mapping against measured frontier reversion at low strain: tier-0 conventions are obeyed (2.3–1.5%), the tier-2 convention defeats every model (76.0%), and tier-1 conventions fall between or split by model. A weighted logistic of reversion on tier across all eight conventions gives $z = 45.4$ ($p < 10^{-6}$).

4.2 Load: the dose-response, with a matched-transform control

Within the taught-reversal contour domain, we raise difficulty (polynomial degree 3–4, 5–8 edges) and measure reversion as competence strain rises. Reversion on the orientation convention climbs monotonically—gpt-oss 1.0%→5.0% at 7% strain, GPT-5 1.0%→13.7% at 24%, Gemma 5.3%→35.8% at 49% (retest mean 34.4%, SD 6.4pp)—while the `wrk` cell, which demands the *identical* negation of the same variables, stays at 0.1–0.3% (Figure 2; weighted-logistic $z = 18.1$).

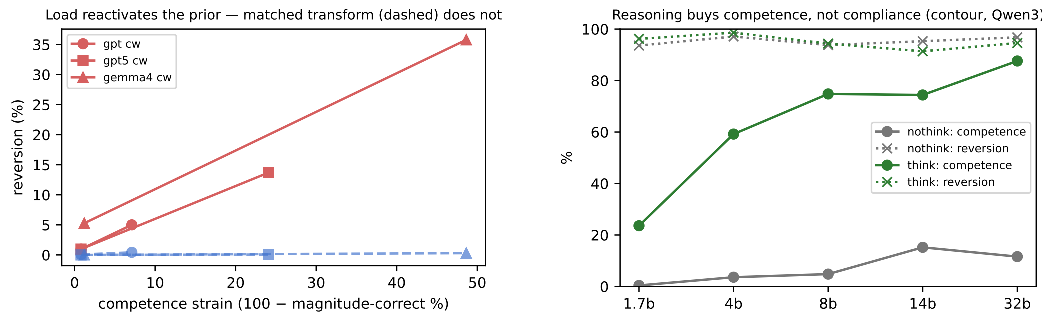

Figure 2: Left (**Fig. 2**): dose-response of orientation reversion vs. strain; dashed lines are the transform-matched `wrk` control. Right (**Fig. 3**): the reasoning toggle lifts competence dramatically while reversion stays pinned (Qwen3, five scales).

Table 2: Reasoning toggle on contour (Qwen3): magnitude-competence % / orientation reversion %.

	no-think		think	
Scale	Comp. %	Rev. %	Comp. %	Rev. %
1.7b	0	93.6	24	96.2
4b	4	97.1	59	98.6
8b	5	93.8	75	94.3
14b	15	95.3	74	91.4
32b	12	96.8	88	94.6

Load does not cause generic instruction loss; it selectively reactivates the trained default. Survivorship in the hardened tier is conservative: rows lost to truncation are the most-strained instances.

4.3 Capability: compliance is frontier-emergent

Mid-scale models revert even on tier-0 conventions frontier models obey: the Qwen3 ladder reverts 91–99% on contour orientation at every scale and both reasoning modes (Table 2), and DeepSeek-V3, Llama-3.3-70B, and Qwen2.5-72B revert 92–94%—though all show large `cw>wrk` gaps (e.g., 92% vs. 35%), so convention identity matters even below the competence floor. Following a taught reversal under load is itself an emergent capability.

5 What does not fix it

Reasoning. Toggling Qwen3's native reasoning mode raises contour magnitude-competence from 0–15% to 24–88% across five scales while moving orientation reversion by 2–4 points (Table 2, Figure 2)—replicating, in a second domain, the dissociation we first observed in circuits. Test-time compute purchases competence, not compliance.

Demonstrations. We prepend one or three fully worked, solver-verified examples of the counter-clockwise convention (held-out physics, different topologies) to the same circuit instances. Gemma declines shallowly (77.9%→71.4%→64.3%, ∼4.5pp per example); gpt-oss (74.2%→73.6%) and GPT-5 (70.6%→75.1%→73.2%) do not move (Figure 3). The prior is decode-dominant: showing the model exactly the behavior required, with verified answers, does not override it. The mitigation practitioners would reach for first does not meaningfully work.

6 Pre-registered predictions, both refuted

Two predictions were committed to the project's version history before the corresponding data existed. **(P1)** We predicted left-handed cross products would behave like counter-clockwise mesh currents (an untrained reversal) and revert heavily. Refuted: across three difficulty tiers and ∼3,700

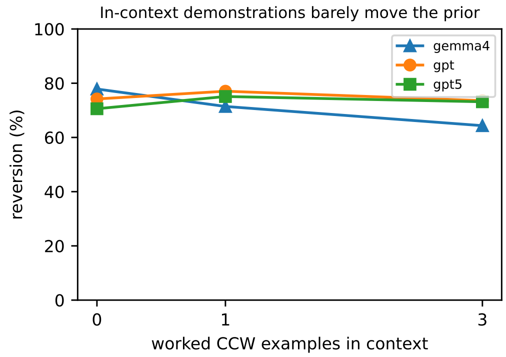

Figure 3: Reversion vs. number of in-context worked counter-clockwise examples.

diagnostic variables, frontier reversion is 0.0–0.6% (0/1231 for Gemma at 95% competence). The left-hand rule is taught—as an error-contrast—in essentially every introductory text; the refutation is what forced the ordinal pedagogy coding of Section 4.1. **(P2)** We predicted in-context demonstrations would collapse reversion if it stems from missing training text. Refuted (Section 5). We report both because a framework that only ever confirms itself is not measuring anything; the surviving claims are the ones these refutations disciplined.

7 Related work

Instruction-following benchmarks measure compliance with formatting and content constraints (Zhou et al., 2023); counter-default instructions have been studied as "inverse" instructions (Stephan et al., 2025) and counterfactual task variants (Wu et al., 2024). Our contribution is orthogonal machinery: sign-exact oracles that separate compliance from competence within one response, matched-transform controls, and pedagogy-tiered conventions. Sycophantic deference to user framing (Sharma et al., 2024) is related but our effect persists with no user stance to defer to. Reasoning-scaling work (Muennighoff et al., 2025; DeepSeek-AI, 2025) measures capability gains from test-time compute; we show those gains do not transfer to convention compliance. The benchmark extends a published single-domain probe (Ravishankara, 2026a), and follows deterministic-evaluation practice from engineering-plot benchmarks (Ravishankara, 2026b); canary-based contamination protection follows Srivastava et al. (2022). Statistical machinery uses exact McNemar tests (McNemar, 1947) and Wilson intervals (Wilson, 1927).

8 Discussion

Why it matters. These failures are silent (magnitudes correct), directed (always toward the trained frame), load-sensitive (worst exactly when problems are hard), and prompt-robust (demonstrations do not help). In engineering workflows a sign is the difference between stable and unstable, tension and compression, charge and discharge. Evaluations that grade magnitudes, final answers modulo sign, or use LLM judges will systematically miss this class.

Limitations. The tier-2 pole is occupied by one convention; our search for a second (P1) instead found a tier-1 case, and we report the gradient rather than a binary law. Strain could not be induced in the vectors domain (95–100% competence at all tiers), so the taught-reversal×high-strain cell is measured only in contour. Pedagogy tiers are coded from a documented but necessarily finite textbook audit. Serving stacks are mixed across the panel (pinned and recorded); hardened-tier truncation censors the most-strained rows conservatively.

Release. The benchmark regenerates byte-identically from seeds; we release all generators, oracles, validators, and analysis code publicly, and the frozen instances and model outputs under gated access with embedded canaries—because a benchmark that measures a training-frequency effect inverts, rather than degrades, under contamination.

References

DeepSeek-AI. Deepseek-r1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv:2501.12948*, 2025.

Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947.

Niklas Muennighoff et al. s1: Simple test-time scaling. In *arXiv:2501.19393*, 2025.

M. Ravishankara. Circuchain: Disentangling competence and compliance in LLM circuit analysis. In *Proc. IEEE SoutheastCon*, 2026a. arXiv:2602.15037.

M. Ravishankara. Plotchain: Deterministic checkpointed evaluation of multimodal LLMs on engineering plot reading. In *Proc. IEEE SoutheastCon*, 2026b. arXiv:2602.13232.

Mrinank Sharma et al. Towards understanding sycophancy in language models. In *Proc. ICLR*, 2024.

Aarohi Srivastava et al. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. *arXiv:2206.04615*, 2022.

M. Stephan et al. Inverse IFEval: Can LLMs unlearn stubborn training conventions to follow real instructions? In *arXiv:2509.04292*, 2025.

Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *JASA*, 22: 209–212, 1927.

Zhaofeng Wu, Linlu Qiu, Alexis Ross, Ekin Akyürek, Boyuan Chen, Bailin Wang, Najoung Kim, Jacob Andreas, and Yoon Kim. Reasoning or reciting? exploring the capabilities and limitations of language models through counterfactual tasks. In *Proc. NAACL*, 2024.

Jeffrey Zhou, Tianjian Lu, Swaroop Mishra, Siddhartha Brahma, Sujoy Basu, Yi Luan, Denny Zhou, and Le Hou. Instruction-following evaluation for large language models. In *arXiv:2311.07911*, 2023.

A Reproducibility

All datasets are deterministic functions of (config, seed); regeneration commands, manifests with canary GUIDs, transform-validation reports (125/125 per domain), the complete statistical freeze (`paper_stats.json`), and per-row provenance (model id, serving stack, quantization, sampling parameters) accompany the code. No LLM is involved at any point in generation, verification, grading, or analysis.